

Research Note

thinkselkl — the premise condition

anchor-KL — the forgetting control

Two components that look like footnotes in the campaign tables and are probably the most transferable results in the tree: the gentlest data channel (thinking-selective) with the exact mechanical reason its pure version failed, and the only regularizer that cut forgetting to 0.25 nats at zero recall cost. All plots generated from the runs' metrics.jsonl.

1. thinkselkl — what it is and why it failed for a good reason

runs lw_n_thinksel* · findings C2-18, C2-32 · weight anatomy '0.8-band'

THE FAMILY. "thinksel" = thinking-selective: the training data's privileged block is the teacher's own

visible <think> trace (not a RAG passage), and only trace-selected items are kept. Four runs:

lw_n_thinksel_0p6b_rag	hybrid twin (task-CE readout)	- CER 0.037 / line-exact 0.953, dCE +0.059
lw_n_thinksel_s43_0p6b_rag	seed-43 twin	- CER 0.041 / 0.949, dCE +0.052
lw_n_thinksel_1p7b_rag	1.7B twin	- CER 0.065 / 0.916, dCE +0.168
lw_n_thinkselkl_0p6b_rag	THE PURE ARM (teacher-KL)	- CER 0.838 / 0.006, dCE -0.023

THE HEADLINE NOBODY ADVERTISED. The thinking channel is the gentlest data channel in the entire project: forgetting +0.05-0.06 nats at 0.6B - five to ten times below every other reciting recipe - with a POSITIVE benchmark delta (+0.5, the only arm above zero) and worst-category damage +0.18, tied with the best. This gentleness replicates across seeds and survives (attenuated) at 1.7B. Its noisy axis is intrusion (5.0% to 17.5% across seeds at n=40; 7.5% at n=200 - under threshold).

WHY THE PURE VERSION COLLAPSED (finding C2-32, the premise condition). thinkselkl recall fell to 0.838 while its hybrid twin sat at 0.037. The diagnosis is mechanical, not statistical:

- In RAG mode, the cache builder ALWAYS premise-checks the teacher: the passage is in the teacher's training-sequence context, so at answer positions the teacher's distribution is SHARP (96.8% teacher-label proximity, measured).
- In thinking mode, the teacher's training-sequence context is its own generated trace - the passage was present only at harvest generation (rag_in_prompt), never in the cached training sequence. Where the trace fails to carry the answer verses, the teacher's distribution at answer positions is DIFFUSE. KL toward a diffuse teacher is a null training signal: there is nothing to transmit.
- Hence: the storage pathway still writes (the weight profile shows a mild, window-tilted deposit peaking at L21), the destruction battery is trivially CLEAN (worst category small, nothing behavioral installed), and the forgetting drift is NEGATIVE (-0.02 nats): an arm that learned almost nothing behavioral and displaced nothing general.

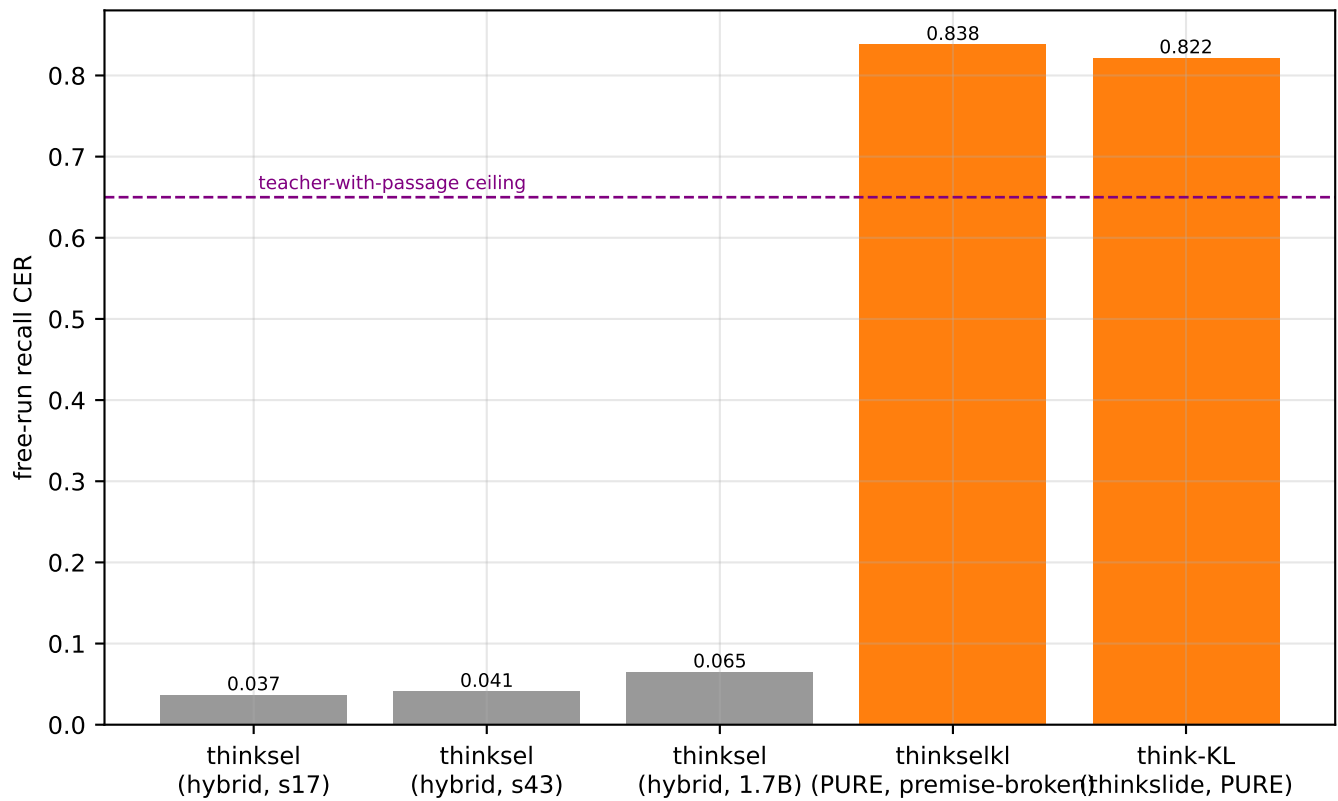
THE LAW IT SHARPENS. teacher_kl transmits only what the teacher knows IN CONTEXT. Purity of the objective is not enough; the teacher must be a premise-checked sharp reader at every training position.

The RAG arms satisfied this by construction; the thinking arms never ran the check.

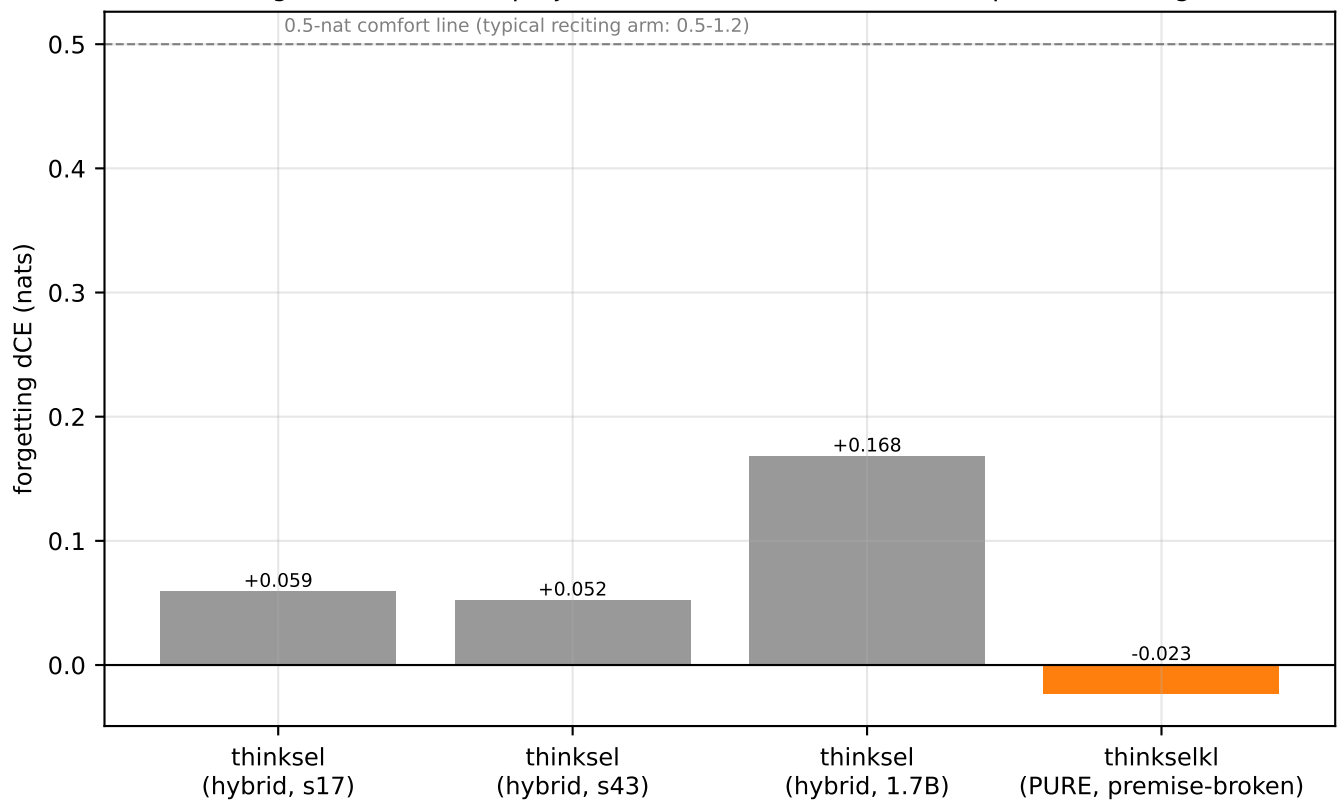
THE FIX (queued, layerwise C3 item 2). Carry the passage inside the thinking teacher's training sequence too (teacher sees passage + trace; student sees neither), or premise-gate per item (drop items where the teacher's answer-position entropy exceeds a threshold). Prediction, worth registering: premise-gated thinkselkl lands in the plateau band (~0.8 - the last-3% law still binds at 0.6B) but inherits the channel's gentleness; at 14B+ it should close entirely, giving the cleanest recipe in the program: gentle data + pure objective + scale.

2. The thinksel family — recall, forgetting, and the premise split

Recall: hybrid twins recite (gray, label-driven); pure arms sit on the plateau (orange)
— the split is the premise condition, not the channel

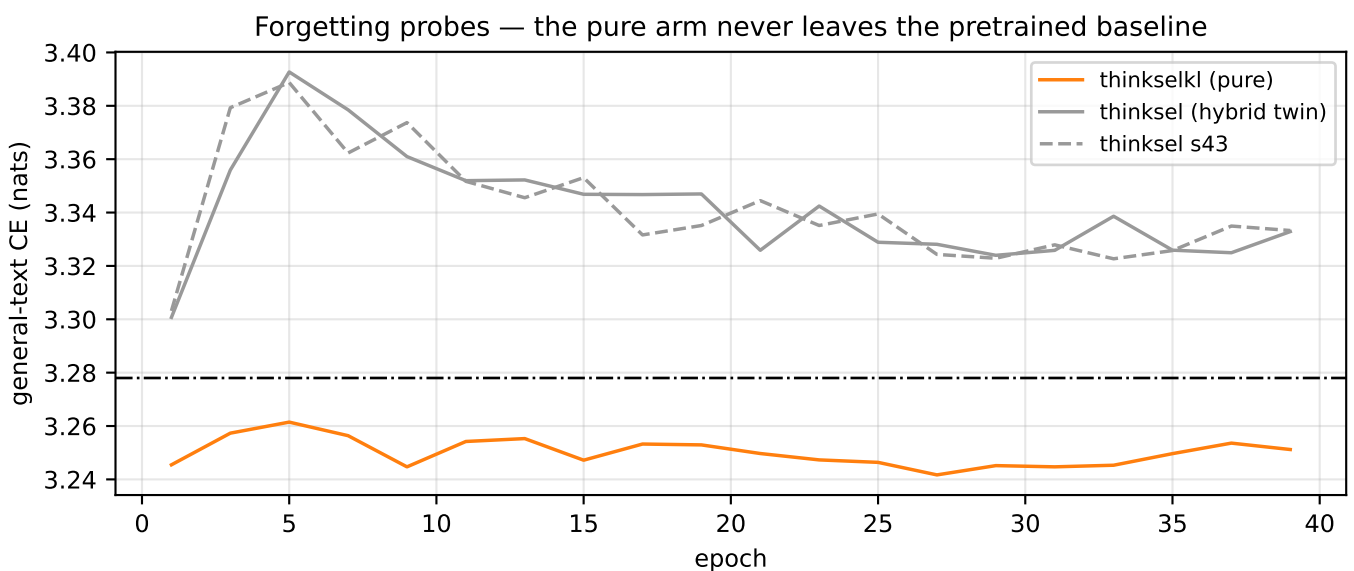
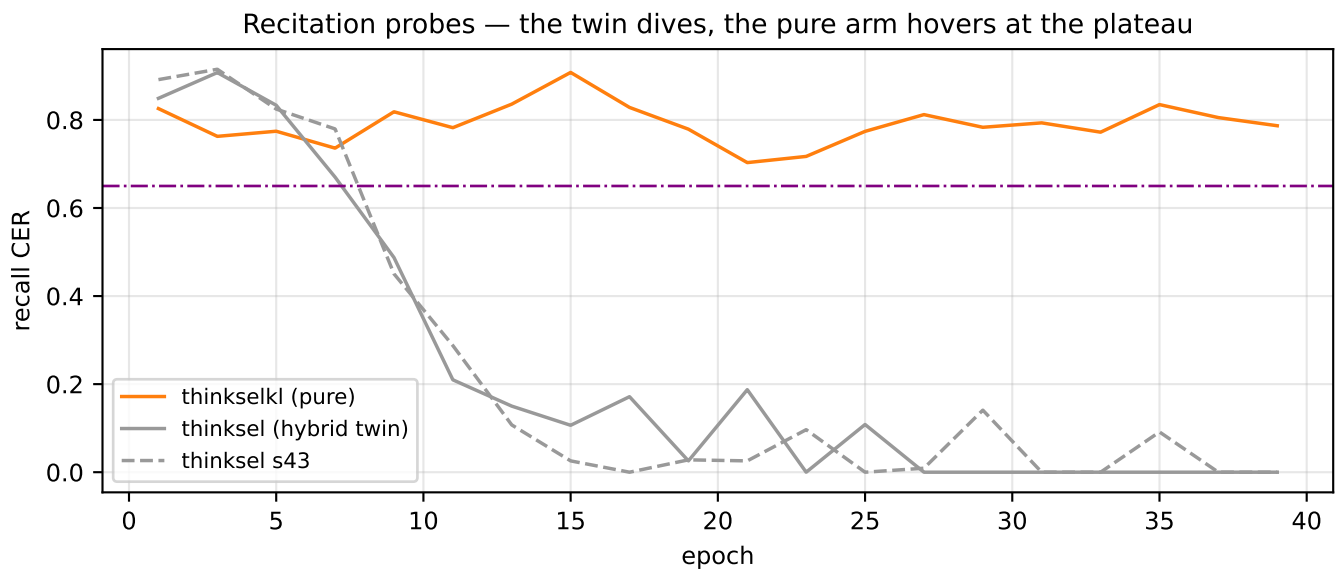
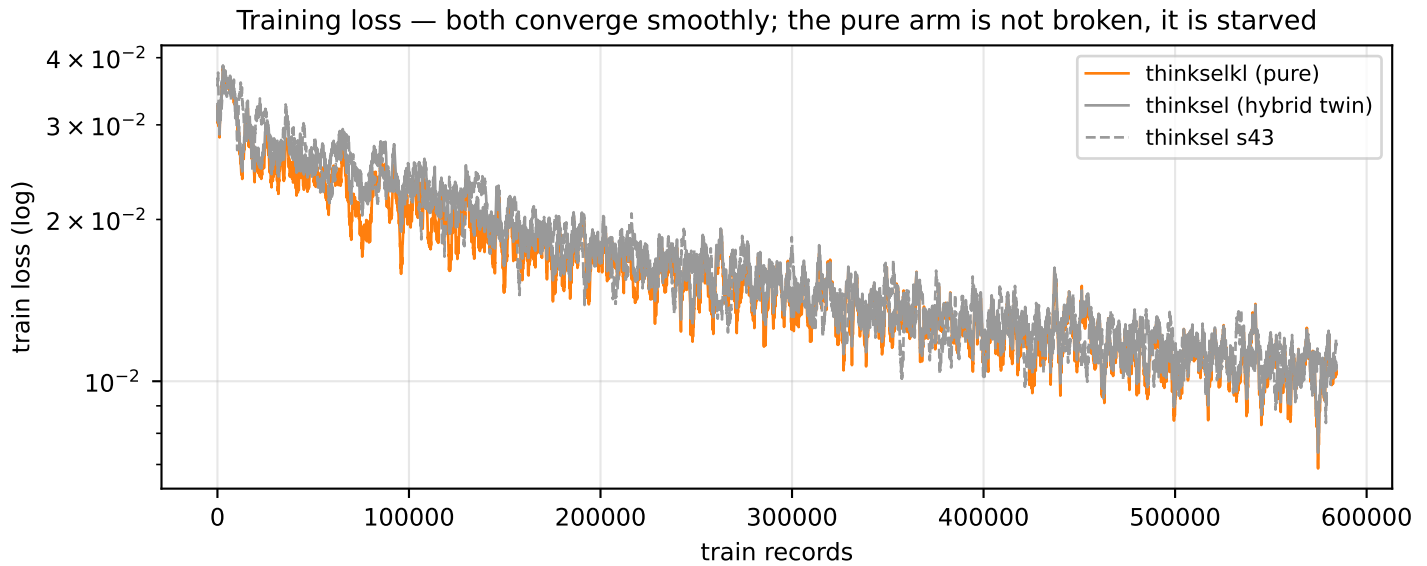


Forgetting: the channel's headline — +0.05 nats WITH recall (gray), and the only negative drift in the project (thinkselkl, learned little, displaced nothing)



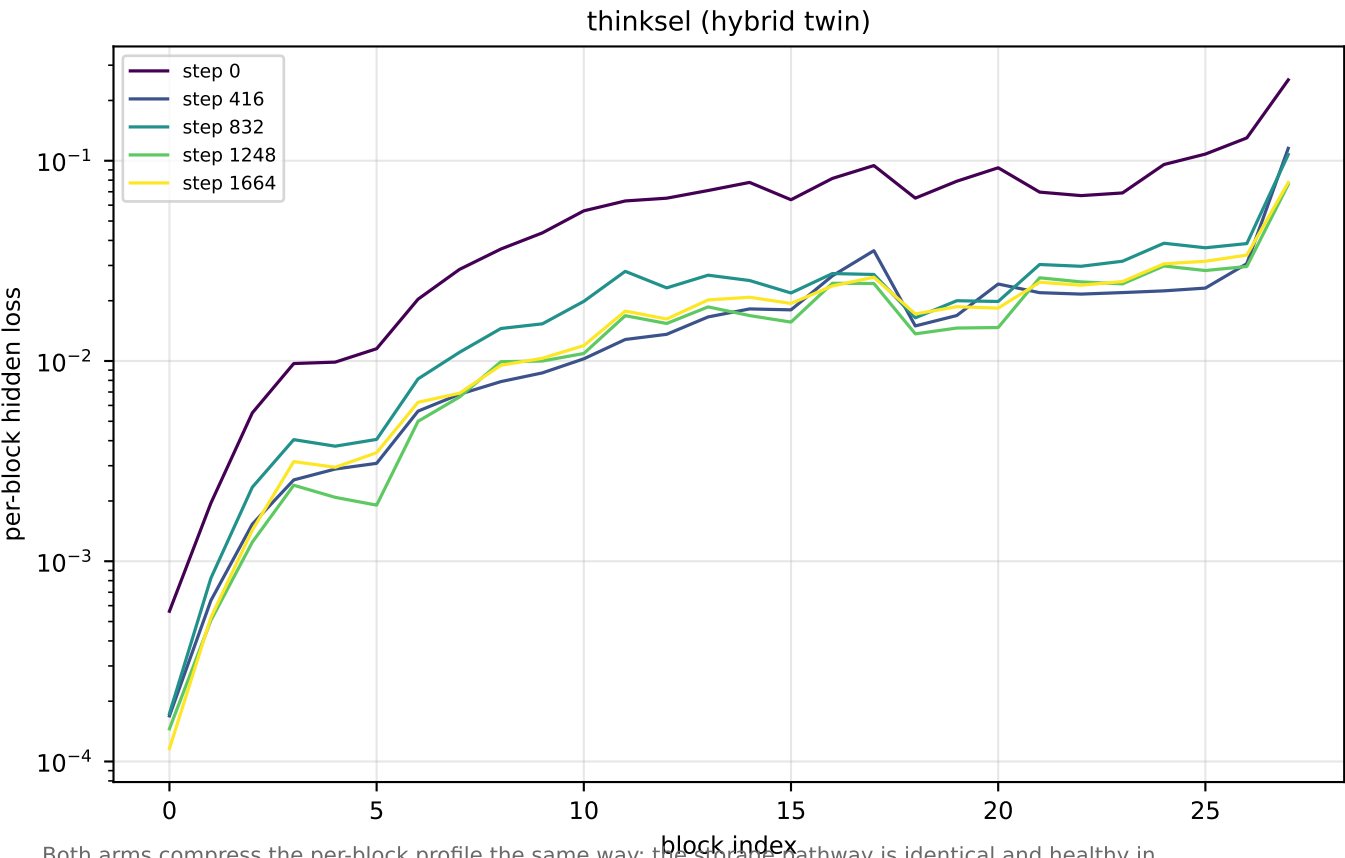
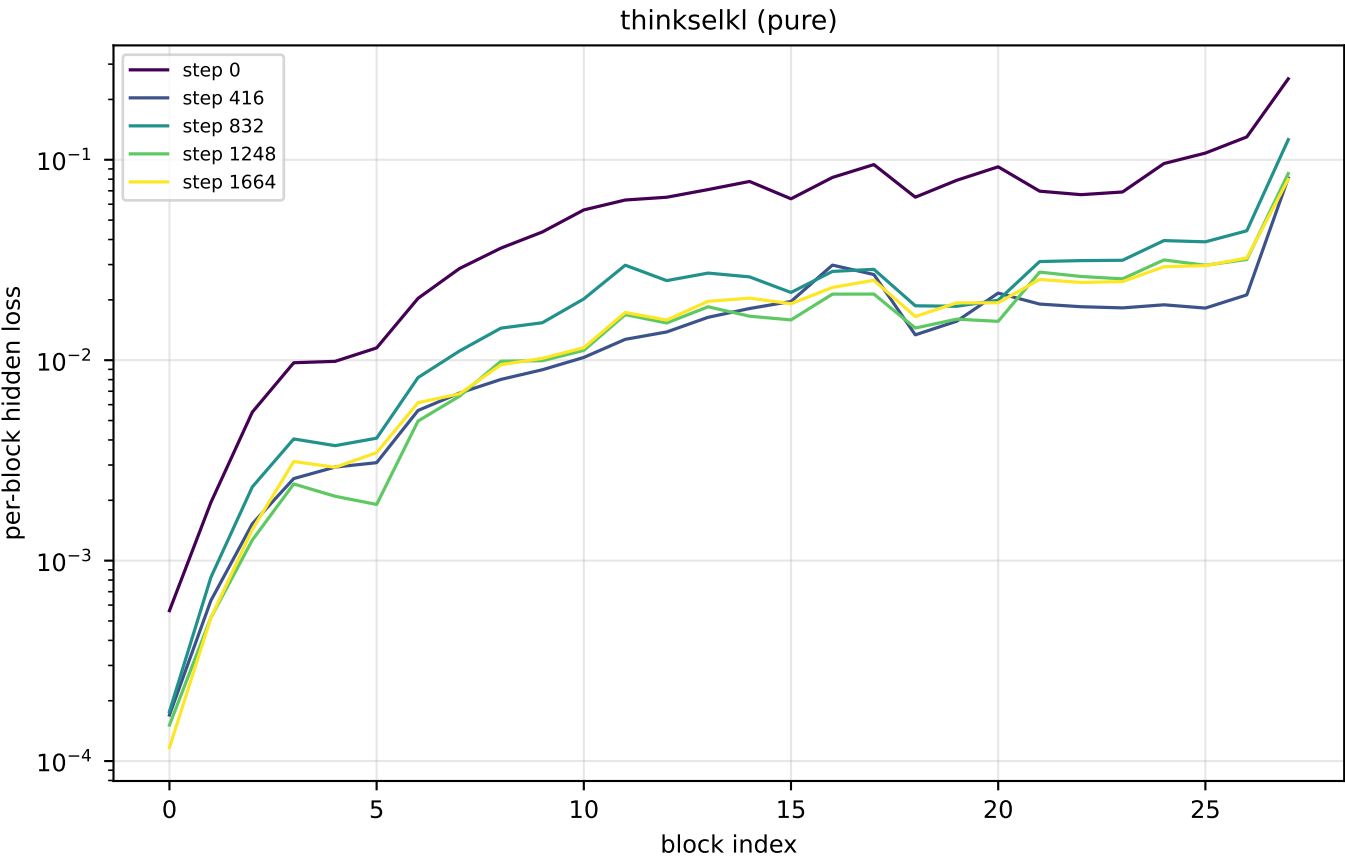
Numbers from runs/results.md. Note what the two panels say together: the thinking-selective channel produces the project's gentlest reciting arms AND its gentlest non-reciting arm. The channel is gentle unconditionally; whether it recites is decided entirely by whether the readout signal carries information (labels always do; teacher-KL only when the teacher is premise-checked).

3. thinkselkl vs its hybrid twin — dynamics from metrics.jsonl



The three panels are the premise condition in raw data: identical data, identical storage loss, identical schedule; the only difference is the readout target (teacher-KL vs reference labels). The pure arm's flat recall at a healthy train loss is the signature of a NULL readout signal, not of failed optimization.

4. thinkselkl — per-block hidden-loss profiles (storage still writes)



Both arms compress the per-block profile the same way: the storage pathway is identical and healthy in the pure arm (weight anatomy: mild, window-tilted deposit peaking at L21). What the pure arm lacks is exclusively the readout sharpening — consistent with a null KL signal at answer positions rather than any storage deficiency.

5. anchor-KL — the forgetting control, and the Goodhart it survives

runs `lw_k_anchor*` · `lw_m_anchordiv*` · findings C2-2, C2-3

WHAT IT IS. anchor-KL adds a regularizer: a KL between the training model and the FROZEN BASE MODEL's own distribution on a small set of unrelated anchor texts. It is teacher-sourced by construction (the 'teacher' here is the untouched base model), so it is admissible in any pure recipe. anchor-CE - the same idea with cross-entropy on fixed fragments - was tried once and recorded as a NEGATIVE (worsens intrusion): even for regularizers, distribution matching beats label matching.

THE FAMILY (all at 0.6B, v2/v4 Machado, matched budgets; carrier recipe is the hybrid final_k8 - the anchor RESULT transfers, the carrier's recall numbers are label-driven and not the point):

<code>lw_k_anchor</code>	anchor-CE variant	dCE +1.93 (the recorded negative)
<code>lw_k_anchorkl</code>	anchor-KL, poetry anchors	dCE +0.50 (halves the no-anchor +1.01 at zero recall cost)
<code>lw_k_final_k8</code>	carrier + poetry anchors	dCE +0.51
<code>lw_m_anchordiv</code>	+ 6 multi-genre anchors	dCE +0.32
<code>lw_m_anchordiv_w1</code>	+ weight 1.0	dCE +0.25 (recall also improved: 0.004)
<code>lw_m_anchordiv_s43</code>	seed replication	dCE +0.33
<code>lw_m_anchordiv_1p7b</code>	1.7B replication	dCE +0.41

FINDING C2-2, ANCHOR-GOODHART. The original anchor file was six POETRY fragments. The destruction battery exposed the fingerprint: poetry protected (+0.37, only category under threshold) while prose/procedural/facts absorbed displaced damage. The C1 claim 'anchor-KL halves neighbor damage' had been measured on poetry probes - where the anchors are. Regularizers protect where anchored.

FINDING C2-3, THE CAUSAL TEST. Same recipe, one change: `anchors_es_v2` = 6 poetry + 6 multi-genre. Every category improved simultaneously (poetry +0.37->+0.27, prose_es +0.81->+0.47, procedural +0.75->+0.37, facts +0.63->+0.48), benchmarks -7.0->-5.5, intrusion 22.5%->12.5%, recall UNCHANGED. Anchor diversity is a free lunch. The w=1.0 sweep then showed stronger pull is also free: dCE +0.25 with the family's best recall.

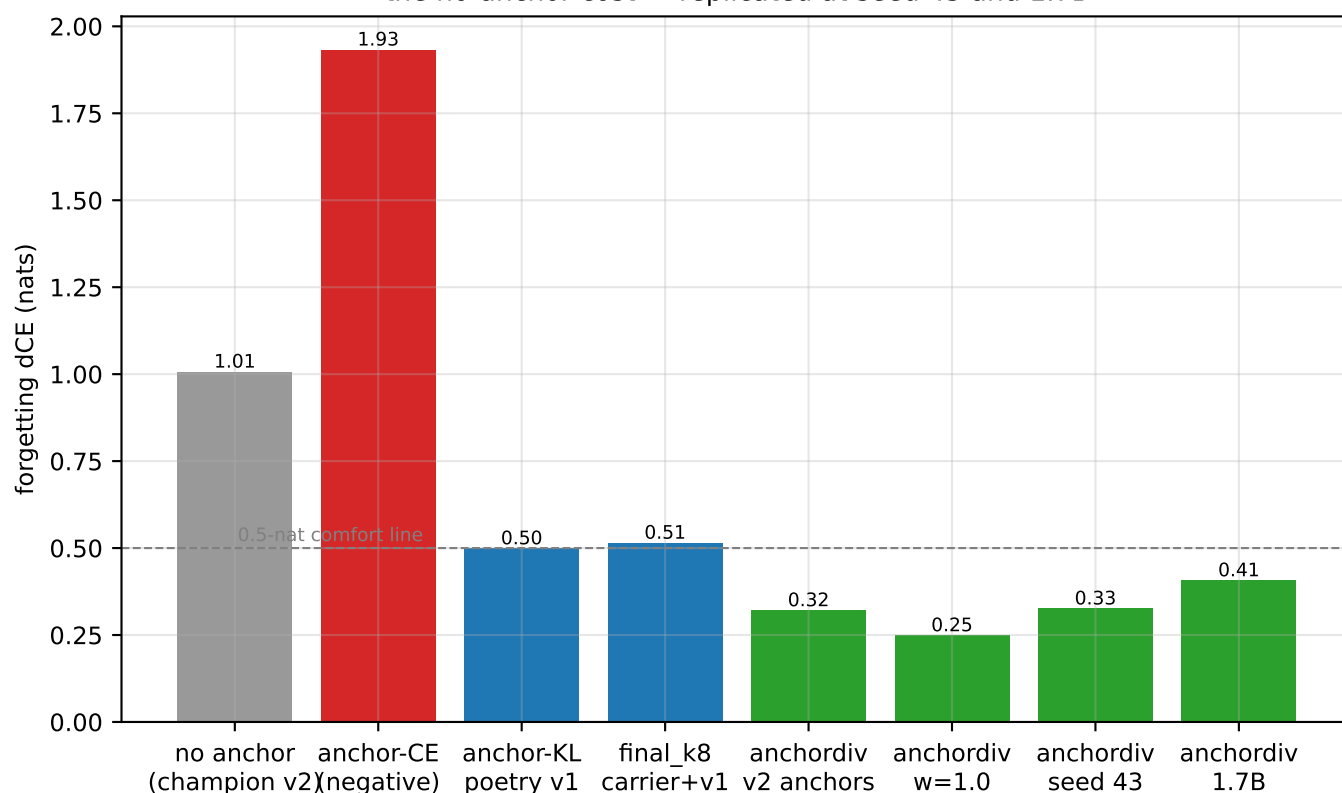
WHAT MAKES THIS THE MOST TRANSFERABLE RESULT IN THE TREE:

- It is a pure-admissible component measured with the project's best forgetting instrumentation.
- Its effect is large (2-4x forgetting reduction), replicated (seed, scale), and mechanistically understood (protection follows the anchor distribution).
- It has NEVER been combined with a pure recipe or with the classic_kd runs. The 14B KD run's +0.35 drift - the largest of the KD slate - is exactly the number diverse-anchor-KL should cut.

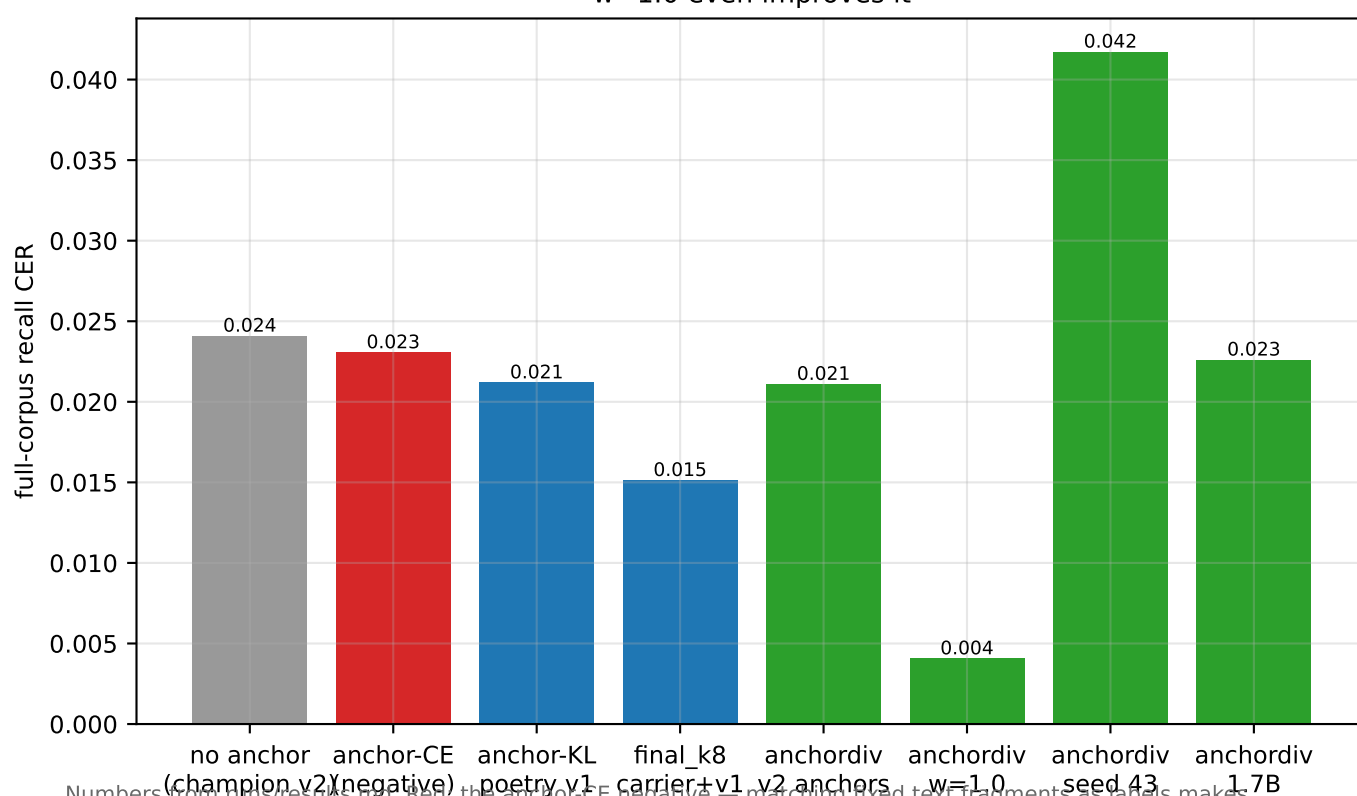
CAVEATS. (a) Carrier taint: all recall numbers in this family are label-driven; only the FORGETTING deltas and battery profiles are the result. (b) The w1 arm's intrusion verdict at n=200 was still in the queue at snapshot. (c) At 1.7B the effect persists but attenuates (+0.41); the anchor set size (6-12 texts) may need to scale with model capacity - untested.

6. The anchor family — forgetting is the result; recall is the carrier's

THE RESULT: anchor-CE hurts (red); anchor-KL halves; diverse anchors + w=1.0 quarter the no-anchor cost — replicated at seed 43 and 1.7B



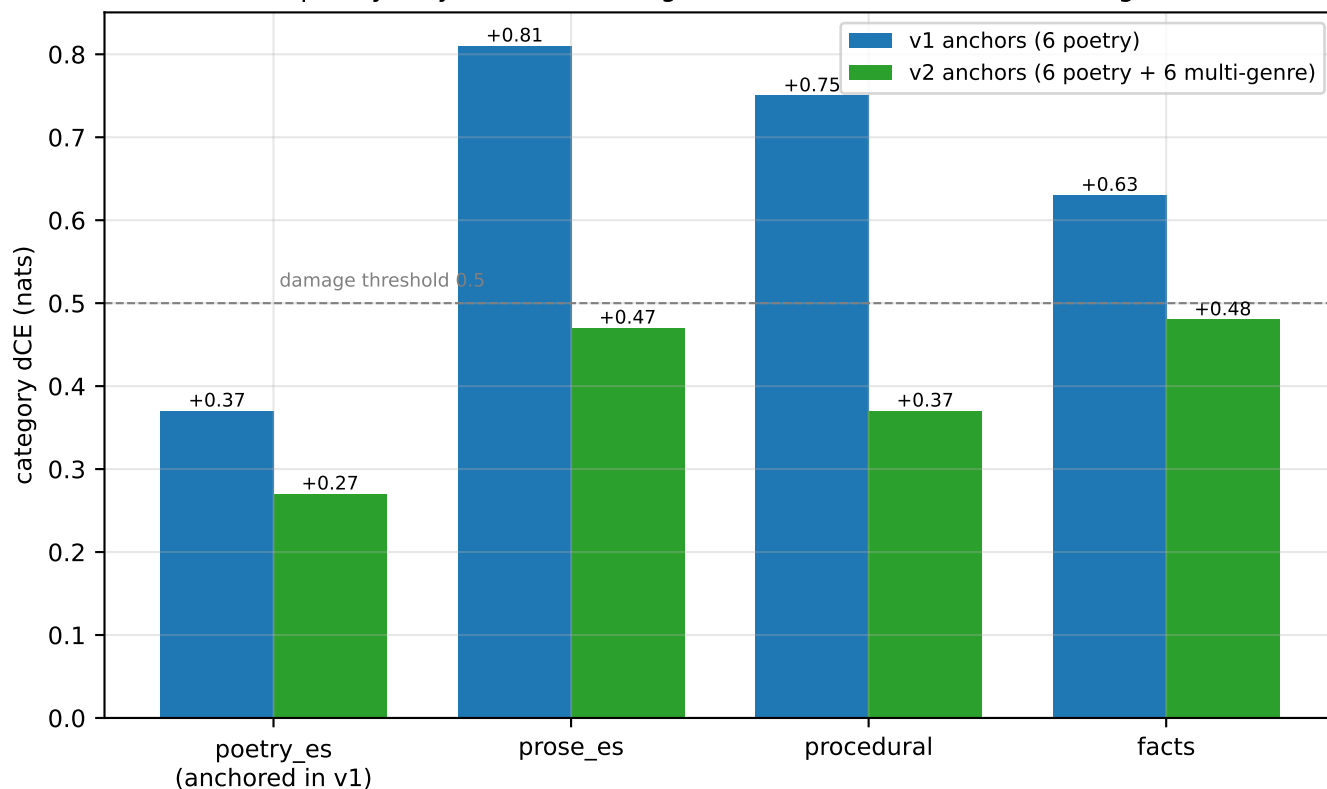
Context only (label-driven carrier): recall is never paid for the protection — w=1.0 even improves it



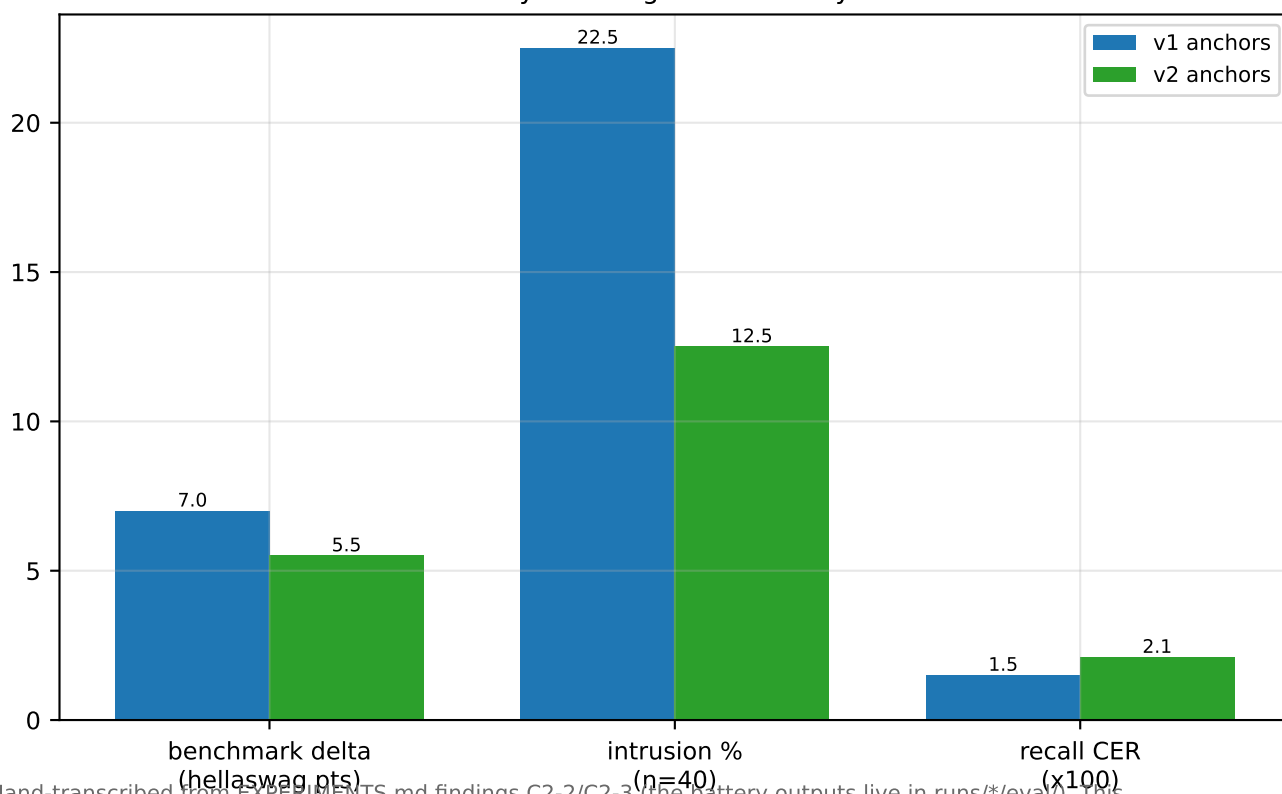
Numbers from runs/results.md. Red: the anchor-CE negative — matching fixed text fragments as labels makes forgetting WORSE (+1.93), the regularizer-level version of the branch's whole task-CE critique. Blue to green: distribution-matching anchors, progressively diversified and strengthened. The green family is the best forgetting control in the project and is pure-admissible: the anchor teacher is the frozen base model itself.

7. Anchor-Goodhart — protection follows the anchor set (finding C2-2/3)

Category-resolved damage, matched recipe and budget: only the anchor file changed.
With poetry-only anchors, damage is DISPLACED into unanchored genres

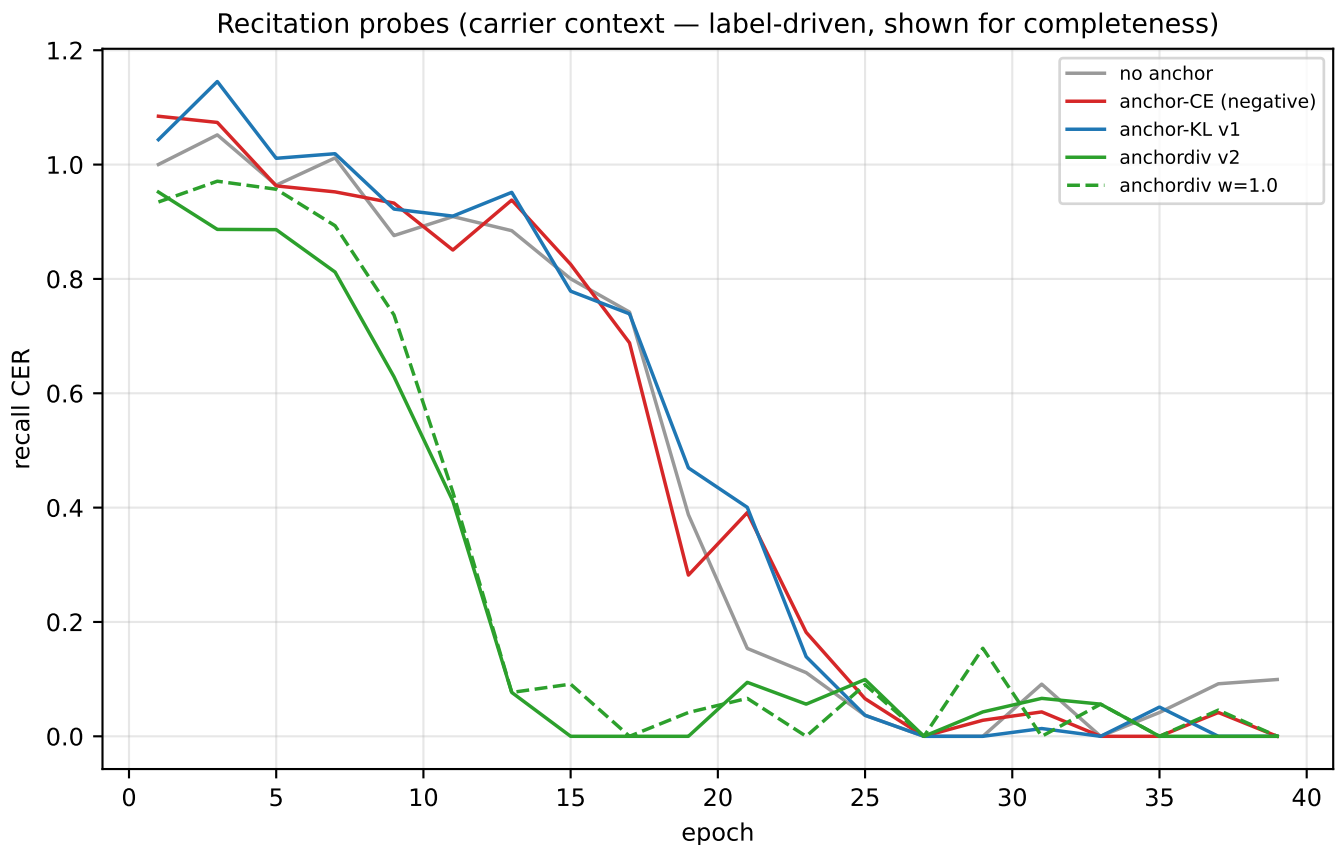


Same comparison, remaining axes: benchmarks recover, intrusion nearly halves, recall statistically unchanged — diversity is a free lunch



Hand-transcribed from EXPERIMENTS.md findings C2-2/C2-3 (the battery outputs live in runs/*/eval/). This is the cleanest causal experiment in Campaign 2: one file changed, every instrument improved. It also retroactively explains Campaign 1's optimistic forgetting numbers: they were measured on poetry probes, where the anchors are.

8. Anchor family dynamics — the control acting in real time



Eval records from each run's metrics.jsonl. The forgetting panel is the note's thesis in one picture: the no-anchor and anchor-CE curves climb away from the pretrained baseline; the anchor-KL curves bend flat; diversity and weight flatten them further. The recall panel shows the price of the control: none.

9. Joint recommendation

one arm, four measured components, decides the program's central question at scale

The two components in this note compose. The proposed arm - assembled entirely from measured pieces, each caveat labeled:

data thinking-selective channel, PREMISE-GATED (C2-32 fix: passage inside the thinking teacher's training sequence, or per-item entropy gate)
storage vocab_mse hidden matching, sliding k=8 window (the safe loss; uniform depth credit)
readout teacher-KL only (pure); expect the plateau at 0.6B - run it at 14B+ where classic_kd shows the plateau dissolving
control anchor-KL, diverse anchors (v2 file or better), weight 1.0

Predicted profile, from the components' independent measurements: forgetting well under +0.3 nats (thinking channel 0.05-0.17 + anchordiv headroom), intrusion under 10% (both components' measured bands), storage full (last-3% law: storage always arrives), recall = the scale question. If the 14B version recites, the program has its Quijote recipe: label-free, bounded backward depth, gentle, anchored. If it plateaus even at 14B, that is a sharp and publishable boundary for label-free layerwise consolidation - either way the arm decides something.

Cost estimate from the family's own runs: the 0.6B version is a ~30-45 min single-GPU run; the 14B LoRA version fits the existing kd harness lanes (~30 GiB, hours-scale). Both fit the standing 12k-item floor discipline.